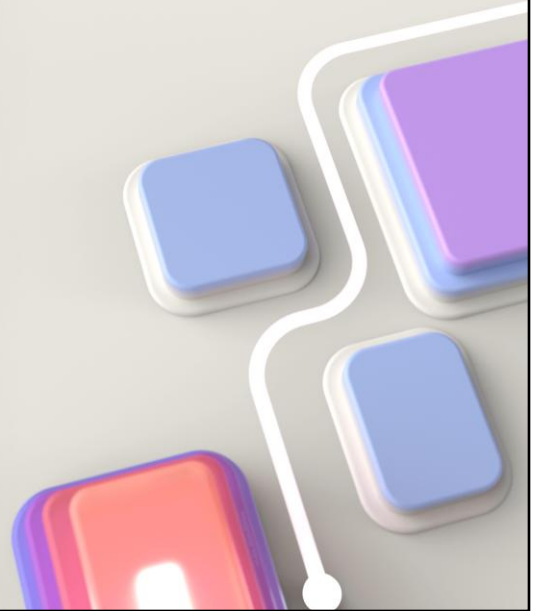


# PL-400T00

## Microsoft Power Platform Developer

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## Supplementary Learning Path 1: Work with Microsoft Dataverse



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This learning path covers the capabilities of Microsoft Dataverse as the foundation for the rest of the course.

The learning path covers creating a data model using Dataverse 's capabilities as the basis for building apps and flows.

This learning path is optional as it is **as these tasks are not the focus of the training/exam per the latest update to Objective Domain.**

## Modules



- Introduction to Microsoft Dataverse
- Manage environments
- Manage customizations with solutions
- Create and manage tables in Microsoft Dataverse

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### Modules

1. Introduction to Microsoft Dataverse
2. Manage environments
3. Manage customizations with solutions
4. Create and manage tables in Microsoft Dataverse
5. Create and manage columns in Microsoft Dataverse
6. Create relationships between tables in Dataverse
7. Create and define calculated and rollup columns in Dataverse
- 8. Define and create business rules in Microsoft Dataverse**
9. Manage security in Microsoft Dataverse

## Modules



- Create and manage columns in Microsoft Dataverse
- Create relationships between tables in Microsoft Dataverse
- Create and define calculated and rollup columns in Microsoft Dataverse
- Define and create business rules in Microsoft Dataverse
- Manage security in Microsoft Dataverse

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### Modules

1. Introduction to Microsoft Dataverse
2. Manage environments
3. Manage customizations with solutions
4. Create and manage tables in Microsoft Dataverse
5. Create and manage columns in Microsoft Dataverse
6. Create relationships between tables in Dataverse
7. Create and define calculated and rollup columns in Dataverse
- 8. Define and create business rules in Microsoft Dataverse**
9. Manage security in Microsoft Dataverse

# Module 1: Introduction to Microsoft Dataverse

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## Introduction to Microsoft Dataverse

In this module we will

- Explain Relational Databases
- Explain what environments, tables, columns, and relationships are in Microsoft Dataverse
- Describe the difference between Microsoft Dataverse and Common Data Model
- Explain use cases and limitations of business rules and process flows

Learning objectives

- Discover Microsoft Dataverse
- Learn about the Common Data Model
- Identify tables, columns, and relationships
- Learn about environments
- Discover business rules

## Learning Objectives

After completing this module, you will be able to:

- 1** Examine the different Power Platform administrative options available.
- 2** Describe security and governance in Power Platform.
- 3** Explore Microsoft Dataverse and how to use it to build business solutions.
- 4** Examine Power Platform connectors.

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# Microsoft Learn module

## Identify foundational components of Microsoft Power Platform

<https://learn.microsoft.com/training/modules/introduction-common-data-service>



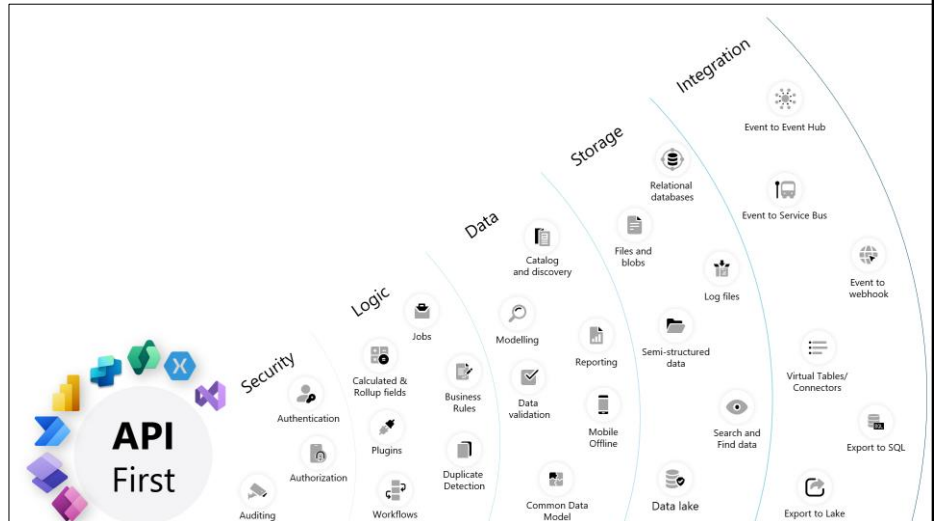
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Link to student courseware on Learn

<https://learn.microsoft.com/training/modules/introduction-common-data-service>

# Dataverse

Microsoft Dataverse is a cloud-based, low-code data service and app platform, that easily structures a variety of data and business logic to support interconnected applications and processes in a secure and compliant manner



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Microsoft Dataverse is a cloud-based, low-code data service and app platform, which allows you to leverage the security and connectivity of Microsoft services. Dataverse connects easily to all aspects of Microsoft Power Platform so that you can fully control, automate, and strengthen your business. With standard tables and columns, as well as the ability to easily define relationships between your data, Dataverse was built for powerful, scalable solutions.

Microsoft Dataverse is a cloud-based solution that easily structures a variety of data and business logic to support interconnected applications and processes in a secure and compliant manner. Managed and maintained by Microsoft, Dataverse is available globally but deployed geographically to comply with your potential data residency. It is not designed for stand-alone use on your servers, so you will need an internet connection to access and use it.

Dataverse is designed to be your central data repository for business data, and you might even be using it already. Behind the scenes, it powers many Microsoft Dynamics 365 solutions such as Field Service, Marketing, Customer Service, and Sales. It is also available as part of Power Apps and Power Automate with native connectivity built right in. The AI Builder and Portals features of Microsoft Power Platform also utilize Dataverse.

Below is a visualization that brings together the many offerings of Microsoft Dataverse.

As you can see, Microsoft Dataverse offers a great deal of functionality. Below is a brief explanation of each category of features.

- **Security:** Dataverse handles authentication with Microsoft Entra ID (Azure Active Directory (Azure AD)) to allow for conditional access and multi-factor authentication. It supports authorization down to the row and



column level and provides rich auditing capabilities.

- **Logic:** Dataverse allows you to easily apply business logic at the data level. Regardless of how a user is interacting with the data, the same rules apply. These rules could be related to duplicate detection, **business rules**, workflows, or more.
- **Data:** Dataverse offers you the control to shape your data, allowing you to discover, model, validate, and report on your data. This control ensures your data looks the way you want regardless of how it is used.
- **Storage:** Dataverse stores your physical data in the Azure cloud. This cloud-based storage removes the burden of worrying about where your data lives or how it scales. These concerns are all handled for you.
- **Integration:** Dataverse connects in different ways to support your business needs. APIs, webhooks, eventing, and data exports give you flexibility to get data in and out.

As you can see, Microsoft Dataverse is a very powerful cloud-based solution for storing and working with your business data. In the following sections, you will look at Microsoft Dataverse from the lens of data storage for Microsoft Power Platform, where you will start your journey. Keep in mind the additional rich capabilities discussed above which you can explore further as your usage increases.

To get started, Microsoft Dataverse lets you create one or many cloud-based instances of a standardized database. The database includes predefined tables and columns which store data commonly found across nearly all organizations and businesses. You can customize and extend what is stored by adding new columns or tables. The ease of setting up a Microsoft Dataverse data store and standardized data model under it simplifies your ability to concentrate your efforts on building solutions without worrying about infrastructure, storage, and data integration. With your data stored in Microsoft Dataverse, there are many different ways to access it. You can work with the data natively with tools such as Power Apps or Power Automate. Or through connectors and APIs you can connect to Microsoft Dataverse from any business solution. With the power of features such as role-based security and business rules you can trust your data is safe no matter how it is accessed.

# Benefits of Microsoft Dataverse

- **Simple to manage** – Both the metadata and data are stored in the cloud. You don't need to worry about the details of how they're stored.
- **Helps to secure data** – Data is stored so that users can see it only if you grant them access. Role-based security allows you to control access to tables for different users within your organization.
- **Access your Dynamics 365 data** – Data from your Dynamics 365 applications is also stored within Dataverse, which allows you to quickly build apps that use your Dynamics 365 data and extend your apps by using Power Apps.
- **Rich metadata** – Data types and relationships are used directly within Power Apps.
- **Logic and validation** – Define calculated columns, business rules, workflows, and business process flows to ensure data quality and drive business processes.
- **Productivity tools** – Tables are available within the add-ins for Microsoft Excel to increase productivity and ensure data accessibility.

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Whether you are building a canvas app or a model-driven app, Dataverse is the ideal data source because it is the foundational data source of Microsoft Power Platform. As a result, you will experience the most functionality, the deepest integrations, the most features, and the best ease-of-use of any available data sources. From simple web-based data design to robust, role-based security, Microsoft Dataverse is a straightforward platform that you can use to begin designing your data structures and helping to keep them safe. Then, with your data in place, you have rich integration capabilities from Power Apps and the rest of Microsoft Power Platform. Additionally, by applying business rules, you can trust that your business integrity will be maintained no matter what tool you use to interact with the data.

Another benefit of using Dataverse is that all of your data is stored in tables. A table is a set of records that is used to store data, similar to how a table stores data within a database. Dataverse includes a base set of standard tables that cover typical scenarios, but you can also create custom tables that are specific to your organization. Standard and custom tables within Dataverse help provide a secure and cloud-based storage option for your data. Tables allow you to create a business-focused definition of your organization's data for use within apps.

Some benefits of using Dataverse and its tables include:

- **Simple to manage** - Both the metadata and data are stored in the cloud. You don't need to worry about the details of how they're stored.
- **Helps to secure data** - Data is stored so that users can see it only if you grant them access. Role-based security allows you to control access to tables for different users within your organization.
- **Access your Dynamics 365 Data** - Data from your Dynamics 365 applications is also stored within the Dataverse, which allows you to quickly build apps that use your Dynamics 365 data and extend your apps by using Power Apps.
- **Rich metadata** - Data types and relationships are used directly within Power Apps.
- **Logic and validation** - Define calculated columns, business rules, workflows, and business process flows to ensure data quality and drive business processes.
- **Productivity tools** - Tables are available within the add-ins for Microsoft Excel to increase productivity and ensure data accessibility.

# Microsoft Dataverse defined

- 1 A Dataverse data store is a single instance of Microsoft Dataverse which stores data in a set of standard and custom data structures called tables.
- 2 A Dataverse data store is scalable and supports both large data sets and complex data models.
- 3 The standard table design in a Microsoft Dataverse data store is based upon an open data model standard called Common Data Model. The structure of a Microsoft Dataverse data store is based upon the definitions and schema in the Common Data Model.
- 4 Dataverse contains comprehensive business logic, processing, integration, and security features that aid the building of business solutions.

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A Dataverse data store is a single instance of Microsoft Dataverse which stores data in a set of standard and custom data structures called tables. A table is a logical set of rows that is used to store data. Rows within a table contain many columns to manage individual pieces of information about a single row.

You can create one or many database instances in Microsoft Dataverse to host data behind your business solutions. Each instance of a Microsoft Dataverse will start with the same set of tables to store data, but you can always extend and customize a Microsoft Dataverse data store to meet specific business needs. This means that you can share business solutions that reference standard tables in Microsoft Dataverse across your organization or with any other organization by using it anywhere in the world.

A Dataverse data store supports large data sets and complex data models. Tables can hold millions of items, and you can extend the storage in each instance of a Microsoft Dataverse data store to four (4) terabytes per instance. The amount of data that is available in your instance of Microsoft Dataverse is based upon the number and type of licenses that are associated with it. Data storage is pooled between all licensed users, so you can allocate storage as needed for each solution that you build. Additional storage can be purchased if you need more storage than what is offered within standard licensing.

**Tip:** Dataverse supports transactional multi-user applications, where quick response to user demand is the priority. It is not intended to be a platform for long running or batch processing.

The standard table design in a Microsoft Dataverse data store is based upon an open data model standard called Common Data Model. Common Data Model is a logical design that includes a set of open-sourced, standardized, extensible data tables and relationships that Microsoft and its partners have published in an industry-wide initiative called the Open Data Initiative. This collection of predefined tables, columns, semantic metadata, and relationships form the basis of the Common Data Model. The structure of a Microsoft Dataverse data store is based upon the definitions and schema in the Common Data Model. The key benefit of using Common Data Model as the basis of a Microsoft Dataverse data store is simplified integration of any solutions that use a Common Data Model schema, because the standard tables of the solution are the same. You will also

be able to take advantage of a rich ecosystem of solutions that vendors have built from using Common Data Model. Best of all, there is practically no limit to how far you can extend a Microsoft Dataverse data store.

Dataverse contains comprehensive business logic, processing, integration, and security features that aid the building of business solutions. These features are covered later in the course.

# Module 2: Manage environments

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## Create and manage environments in Dataverse

In this module we will

- Environments in Microsoft Dataverse
- Create and manage an environment
- Create an instance of a Microsoft Dataverse data store
- Add users and roles within an environment
- Manage settings in an environment

### Learning objectives

- Identify the different environments that are in Dataverse.
- Create and manage environments in Dataverse.
- Create an instance of a data store in Dataverse.
- Add users and security roles within an environment in Dataverse.
- Determine where to manage settings for an environment.

An environment is a space to store, manage, and share your organization's business data that is stored within an instance of a Dataverse data store. You can set up one or many environments, depending on the needs of your organization. This module explores these environments and how you can use them with instances of Dataverse data stores.

## Learning Objectives

After completing this module, you will be able to:

- 1 Identify the different types of environments.
- 2 Create and manage environments.
- 3 Create an instance of Dataverse in an environment.
- 4 Add users and security roles within an environment with Dataverse.
- 5 Determine where to manage settings for an environment.

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# Microsoft Learn module

## Create and manage environments in Dataverse

<https://learn.microsoft.com/training/modules/create-manage-environments/>



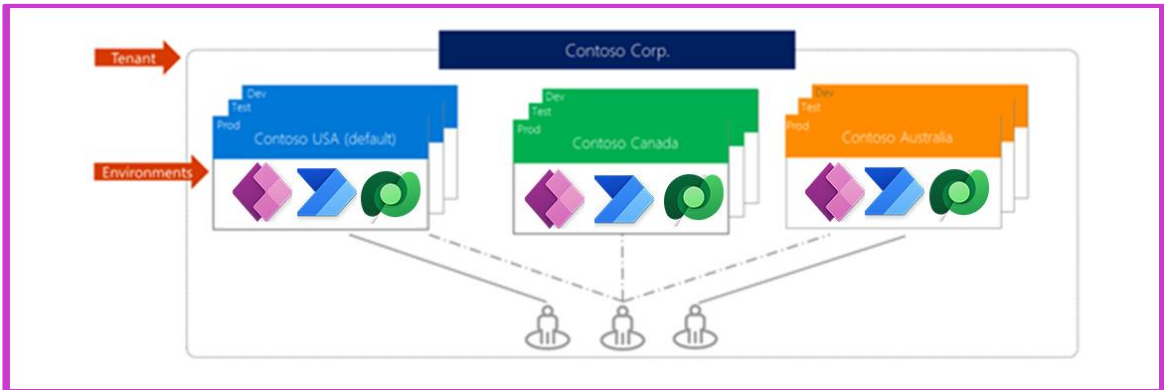
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Link to student courseware on Learn

<https://learn.microsoft.com/training/modules/create-manage-environments/>

# Environments in Power Platform

Power Platform environments are used to store, manage, and share your organization's business data, apps, flows, and users.



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Environments are used to store, manage, and share your organization's business data, apps, and flows in Microsoft Power Platform. Each environment allows you to provision one Microsoft Dataverse data store for use within that environment. Microsoft Dataverse environments allow you to manage user access, security settings, and the storage that is associated with that database.

An environment is a way to create and manage an instance of a Dataverse data store.

Each environment is created under a Microsoft Entra ID tenant, and its resources can only be accessed by users within that tenant. An environment is also bound to a geographic location, like the United States. When you create a Microsoft Dataverse data store in an environment, that database is created within datacenters in that geographic location. Any items that you create in that environment (including connections, gateways, flows that are using Power Automate, and more) are also bound to their environment's location. Contoso tenant to environment map

You can create more than one environment to manage solution development and data storage by setting up one environment for development, another for testing, and another for production use. Also, you can set up an environment based on a geographical location. For example, you might set up an environment for Europe and another for Asia. Each of these environments will have zero or only one instance of Microsoft Dataverse.



## Demo: Create a Power Platform environment

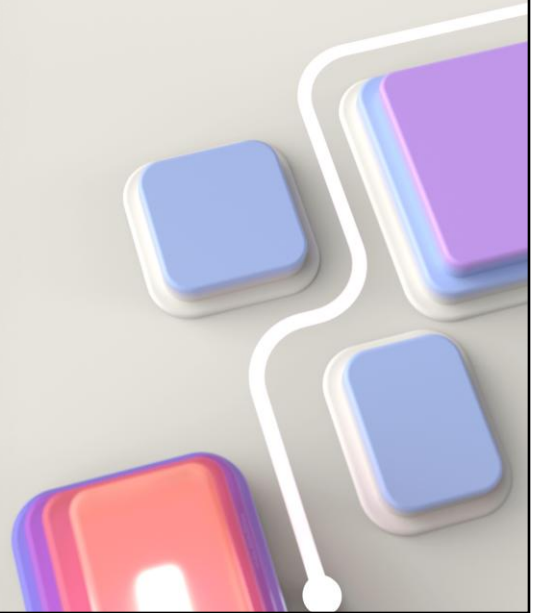
DEMO

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Create a Developer environment with a Dataverse data store

# Module 3: Manage customizations with solutions

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## Manage customizations with solutions

In this module we will

- Solutions
- Publishers
- Adding components to solutions
- Demo - Create a Microsoft Dataverse table

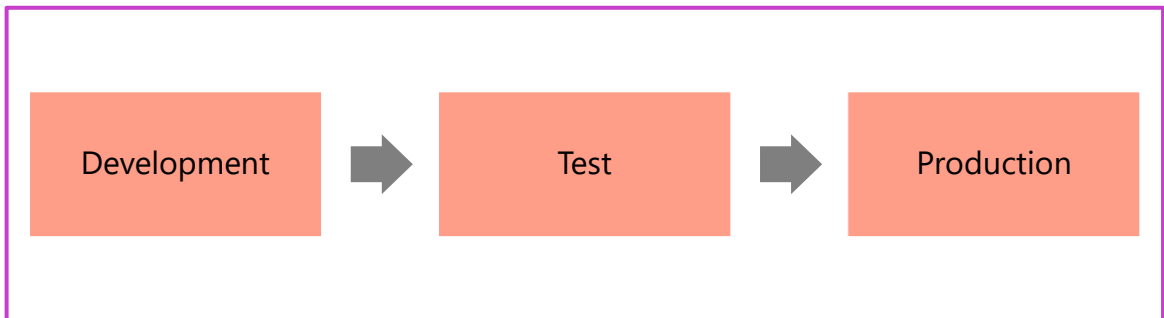
**Note:** There is no Microsoft Learn module for this module until later in the course. This is additional ILT content to introduce the concept of solutions as will be using solutions in the practice labs

Learning objectives

- Unmanaged Solutions

## Solutions

- Solutions are containers of Power Platform components
- Transport customizations from one environment to another
- Part of an overall strategy for application lifecycle management (ALM)



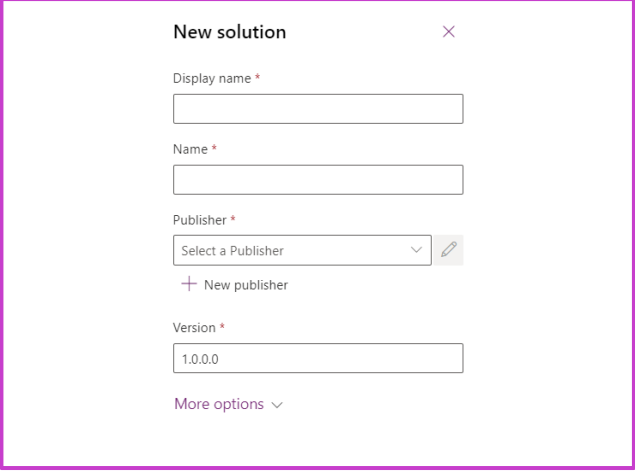
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Just Unmanaged solutions for now

Managed solutions and ALM are covered towards the end of the course

## Create solution

- **Display name** used in Maker portal
- **Name** is the schema name and cannot be changed and cannot have spaces
- **Publisher** defines the prefix for new components
- **Version** number to track updates



The screenshot shows a 'New solution' dialog box with a close button (X) in the top right corner. It contains the following fields and options:

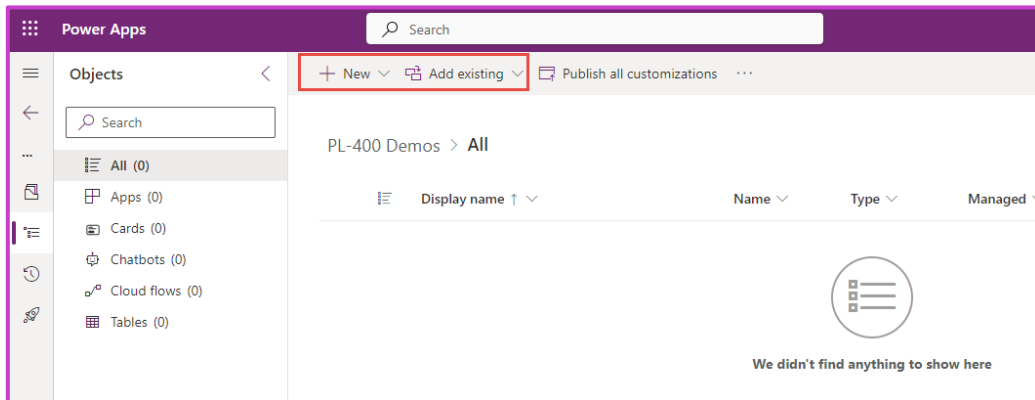
- Display name \***: A text input field.
- Name \***: A text input field.
- Publisher \***: A dropdown menu with the text 'Select a Publisher' and a downward arrow, followed by an edit icon (pencil).
- + New publisher**: A link to add a new publisher.
- Version \***: A text input field with the value '1.0.0.0'.
- More options**: A link with a downward arrow.

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Create in Maker portal

# Components in solutions

- Create new components in solutions
- Add existing components to solutions



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## Demo: Create a solution

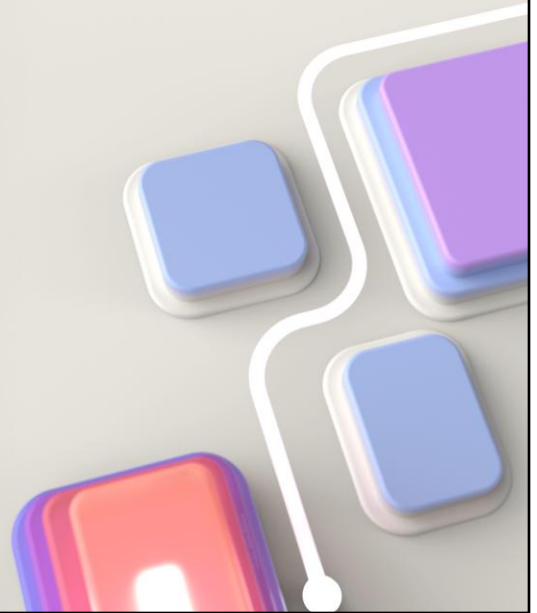
DEMO

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Create a Publisher and a Solution  
Import Solution

# Module 4: Create and manage tables in Microsoft Dataverse

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## Create tables in Dataverse

Data within Dataverse is stored within a set of records called tables.

In this module we will

- Table characteristics
- Dual-write vs. virtual tables
- Dataverse auditing
- Demo - Create a Microsoft Dataverse table

## Learning objectives

- Create tables with Dataverse
- Import data into a Dataverse data store

## Learning Objectives

After completing this module, you will be able to:

- 1 Create and import tables with Dataverse
- 2 Creating a custom table.
- 3 Enabling attachments within a table.

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# Microsoft Learn module

## Create tables in Dataverse

<https://learn.microsoft.com/training/modules/get-started-with-powerapps-common-data-service>

## Manage tables in Dataverse

<https://learn.microsoft.com/training/modules/create-manage-entities>



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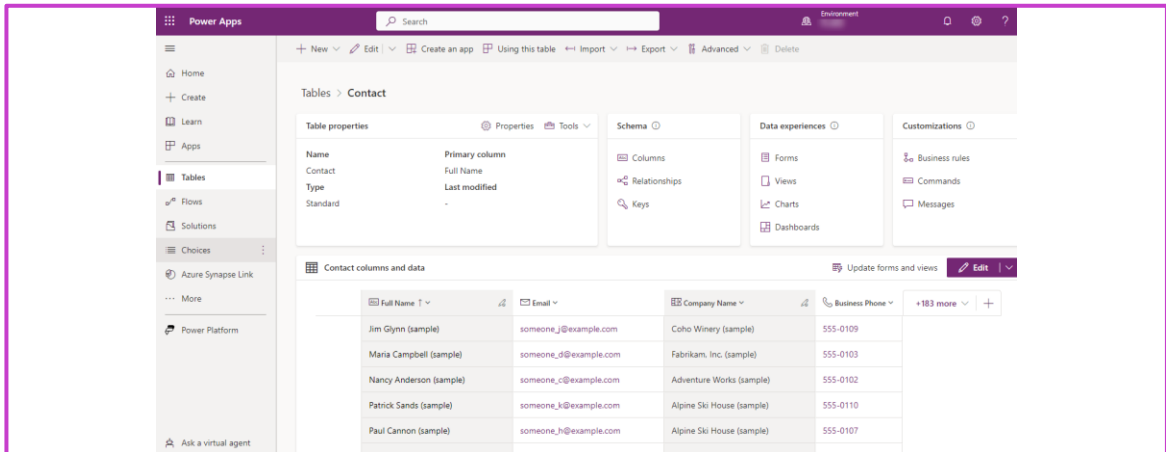
Link to student courseware on Learn

<https://learn.microsoft.com/training/modules/get-started-with-powerapps-common-data-service>

<https://learn.microsoft.com/training/modules/create-manage-entities>

# Identify tables in Microsoft Dataverse

A table is a logical structure containing rows and columns that represents a set of data.



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A table is a logical structure containing rows and columns that represents a set of data.

Data within Dataverse is stored within a set of records called tables. A **table** is a set of records used to store data, similar to how a table stores data within a database.

Dataverse includes a base set of standard tables that support common business scenarios that connect to Dynamics 365 application data. You can also create custom tables specific to your organization and populate them with data that you import from lists in SharePoint, from Excel, or from PowerQuery. App makers can then use Power Apps to build rich applications using this data.

Dynamics 365 applications, such as Dynamics 365 Sales, Service, and Marketing, use Dataverse to store and secure data used by the applications. This means you can build apps using Power Apps and Dataverse directly against your core business data already used within Dynamics 365 without the need for manual integration.

**Note:** Finance and Operations apps currently require the configuration of the [Data Integrator](#) to make your business data from Finance and Operations apps available in Dataverse.

For most organizations, it's a good idea to use the standard tables and columns as they were intended. But to meet your business needs, you can extend the functionality of standard tables by creating one or more custom tables to store information that's unique to your organization.

Data in Dataverse is securely stored so that users can see it only if you grant them access. Role-based security, based on the Dynamics 365 system allows you to control access to tables for different users within your organization.

# Logic and validation

Tables within Dataverse can leverage rich server-side logic and validation to ensure data quality.

- 1 Business rules:** Business rules validate data across multiple columns in a table, and provide warning and error messages, regardless of the app that's used to create the data
- 2 Business process flows:** Business process flows guide users to ensure they enter data consistently and follow the same steps every time. Business process flows are currently supported only for model-driven apps
- 3 Workflows:** Dataverse classic workflows automate business processes without requiring user interaction
- 4 Business logic with code:** Business logic supports advanced developer scenarios that extend the application directly through code

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Tables within Dataverse can leverage rich server-side logic and validation to ensure data quality. You can also reduce repetitive code in each app that creates and uses data within a table.

- **Business rules:** Business rules validate data across multiple columns in a table, and provide warning and error messages, regardless of the app that's used to create the data.
- **Business process flows:** Business process flows guide users to ensure they enter data consistently and follow the same steps every time. Business process flows are currently supported only for model-driven apps.
- **Workflows:** Workflows automate business processes without requiring user interaction.
- **Business logic with code:** Business logic supports advanced developer scenarios that extend the application directly through code.

# Types of tables

The three types of tables you can create are:

## Standard

Create and edit records

## Activity

Used to track interactions on standard tables

## Virtual

Connect to tables in an external data source

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Before creating tables, you should understand that there are different types of tables. Once a custom table is created, these types cannot be changed. The two major divisions are based on table ownership and the tables type.

The types of tables:

- Standard
- Activity
- Virtual
- Elastic (Preview)

**Note:** New tables you create are also known as **Custom**. Custom tables are unmanaged tables that are either imported from an unmanaged solution or are new tables created directly in the Dataverse environment. Any user with appropriate privileges can fully customize these tables.

# Types of tables

## An *activity* is used to track an interaction

- They have time dimensions (start time, stop time, due date, and duration) that help define when the action occurred or will occur.
- They have data (like a subject and description) that helps define the action that the activity represents.
- They can be opened, canceled, or completed. Several sub-status values will be associated with the *Completed* status of an activity to clarify how the activity was completed.
- **All activities are accessed via the same security privileges, you cannot have different levels of access for different activity types.**
- **Appointment:** A commitment representing a time interval that has start/end times and duration.
- **Email:** An activity that's delivered by using email protocols.
- **Fax:** An activity that tracks the call outcome and number of pages for a fax.
- **Letter:** An activity that tracks the delivery of a letter. The activity can store an electronic copy of the letter.
- **Phone Call:** An activity that tracks a telephone call.
- **Recurring Appointment:** The master appointment of a recurring appointment series.
- **Task:** A generic activity representing work that must be done.
- **Custom:** Any custom activity that does not fit with the above out of the box activities.

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An *activity* is an action that a calendar entry can be made for. Activities have these characteristics:

They have time dimensions (start time, stop time, due date, and duration) that help define when the action occurred or will occur.

They have data (like a subject and description) that helps define the action that the activity represents.

They can be opened, canceled, or completed. Several sub-status values will be associated with the *Completed* status of an activity to clarify how the activity was completed.

Activity tables can be owned only by a user or team. They can't be owned by an organization.

All activities share many properties in common such as planned start date and actual start date, planned end date and actual end date, duration, subject and description.

All activity tables are linked to the ActivityPointer table that stores these common columns and lets users view lists of all activities together instead of separate views for each type.

The following default activity tables are available:

- **Appointment:** A commitment representing a time interval that has start/end times and duration.
- **Email:** An activity that's delivered by using email protocols.
- **Fax:** An activity that tracks the call outcome and number of pages for a fax. The activity can optionally store an electronic copy of the document.
- **Letter:** An activity that tracks the delivery of a letter. The activity can store an electronic copy of the letter.
- **Phone Call:** An activity that tracks a telephone call.
- **Recurring Appointment:** The master appointment of a recurring appointment series.
- **Task:** A generic activity representing work that must be done.

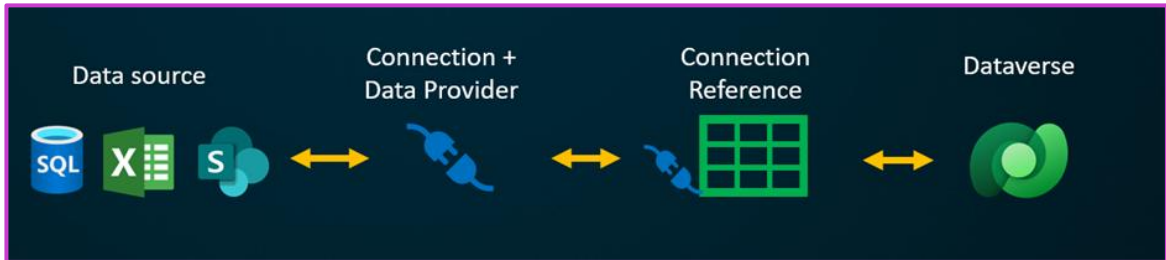
There are two custom activity settings: **Type = Activity** – If this type is selected, the table will be used as an activity

and can be created in the same manner as Letters, Email messages, Phone Calls and so on. After the table is saved, this option is unavailable and cannot be changed and

**Display in Activity Menus** – If the table is defined as an activity this option can be enabled or disabled. In some cases an organization might want activities to be created automatically by workflow processes instead of manually, or only from a sub-grid of a parent record they should relate to. Therefore this option can be cleared so that the activity cannot be created from the navigation bar, command bar, or activity view menus.

# Virtual tables

Virtual tables (also known as virtual entities) enable the integration of data residing in external systems by seamlessly representing that data as tables in Microsoft Dataverse, without replication of data and often without custom coding.



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Virtual tables (also known as virtual entities) enable the integration of data residing in external systems by seamlessly representing that data as tables in Microsoft Dataverse, without replication of data and often without custom coding.

Virtual tables replace previous client-side and server-side approaches to integrating external data, which required customized code and suffered from numerous limitations, including imperfect integration, data duplication, or extensive commitment of development resources. In addition, for administrators and system customizers, the use of virtual tables greatly simplifies administration and configuration.

Find more information on:

<https://learn.microsoft.com/en-us/power-apps/maker/data-platform/create-edit-virtual-entities>

<https://learn.microsoft.com/en-us/power-apps/maker/data-platform/create-virtual-tables-using-connectors>

# Record ownership

When you create a custom table, the options for ownership are:

- 1 User or team owned:** Actions that can be performed on these rows can be controlled at the user level.
- 2 Organization-owned:** Access to the data is controlled at the organization level.

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When you create a custom table, the options for ownership are *User or team owned*, or *Organization-owned*. After a table is created, you can't change the ownership.

- **User or team owned:** Actions that can be performed on these rows can be controlled at the user level.
- **Organization-owned:** Access to the data is controlled at the organization level.

There are four different types of table ownership.

**User or Team** ownership will add a column that is named **Owner** so records for the table can be assigned between Users and Teams that have at least User level read privileges to the table. Internally the owning User or owning Team are stored separately, and the Business Unit they belong to is also stored so there are four lookup columns added that relate to ownership. This use of ownership also means that you can set any of the usual five access levels in Security Roles to control the records that the User can access, depending on where the User and the record are in the Business Unit hierarchy. Most tables that relate to business data are likely to use this level of ownership.

**Organization** ownership means the table does not have an **Owner** column, or the other related lookup columns, and cannot be assigned to a User or Team. Security Roles control whether users have access to all records or no records, there is no separation by Business Units. Tables that do not have many columns and are used as lookups for categorizing other records might be more likely to use Organization level ownership. These include Article, Article Template, Competitor, Currency, and Web Resource.

**Business-owned** - There are several business-owned system tables. These include Business Unit, Calendar, Team, Security Role, and User.

**None** - There are many system tables that don't have an owner, but most of these aren't visible. These mostly consist of intersect tables created to support Many-to-Many relationships or where access to the record is controlled by a parent record. For example, Opportunity Product records must be accessed through a user or team owned Opportunity record.



# Table metadata

Microsoft Dataverse is designed to let you quickly and easily create a data model for your application, based on the tables and the table metadata that you include in your app.

- Display name
- Plural name
- Schema name
- Primary column
- Table type
- Record ownership
- Advanced options

The image shows two side-by-side screenshots of the Microsoft Dataverse configuration interface. The left screenshot, titled 'New table', shows the initial setup form with fields for 'Display name', 'Plural name', 'Description', 'Schema name', 'Type', and 'Record ownership'. A red box highlights the 'Primary column' tab. The right screenshot, titled 'Properties', shows the 'Primary column' configuration page with fields for 'Display name', 'Description', 'Schema name', 'Column requirement', and 'Maximum character count'. A red box highlights the 'Primary column' tab in the 'Properties' section.

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Microsoft Dataverse is designed to let you quickly and easily create a data model for your application, based on the tables and the table metadata that you include in your app.

Tables describe the kinds of data that is stored in the Dataverse data store. Each table corresponds to a database table and each column (also known as an attribute) within a table represents a column in that table.

In Dataverse, metadata (data about data), is a collection of tables. Table metadata is what controls the kinds of rows you can create and what kind of actions can be performed on them. When you use customization tools to create or edit tables, columns, and table relationships, you are editing this metadata.

The apps that your customers use to interact with the data in your environment depend on the table metadata, and they adapt as you customize the metadata.

When a custom table is first created it is available only to users who have either a security role of System Administrator or System Customizer. Other Security Roles must be modified to let other users view and use the new table. These Security Roles can be included in the Solution together with the custom table so that when the Solution is deployed to another system, the roles are ready to be assigned to Users. If the roles already exist in the target system and are being updated, the changes will occur as soon as the Solution is imported.

## Advanced table options

- Enable attachments
- Create activities
- Auditing
- Duplicate detection
- Connections
- Quick create
- Offline

**Note:** Some options cannot be turned off after they have been enabled

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Choose table image

None

+ New image web resource

Color

Enter color code

Refine how data in this table is used and managed. Options marked with <sup>1</sup> can't be turned off if enabled. [Learn more](#)

For this table

☐ Apply duplicate detection rules <sup>1</sup> ⓘ

☐ Audit changes to its data ⓘ

☐ Track changes <sup>1</sup> ⓘ

☐ Leverage quick-create form if available ⓘ

☐ Provide custom help ⓘ

☐ Enable long term retention ⓘ

Help URL

Make this table an option when

☐ Creating a new activity <sup>1</sup> ⓘ

☐ Setting up SharePoint document management ⓘ

☐ Doing a mail merge ⓘ

Rows in this table

☐ Can have connections <sup>1</sup> ⓘ

☐ Appear in search results

☐ Can have a contact email <sup>1</sup> ⓘ

☐ Can be taken offline ⓘ

☐ Have an access team ⓘ

☐ Can be added to a queue <sup>1</sup> ⓘ

Some of the common options that can be enabled for a table.

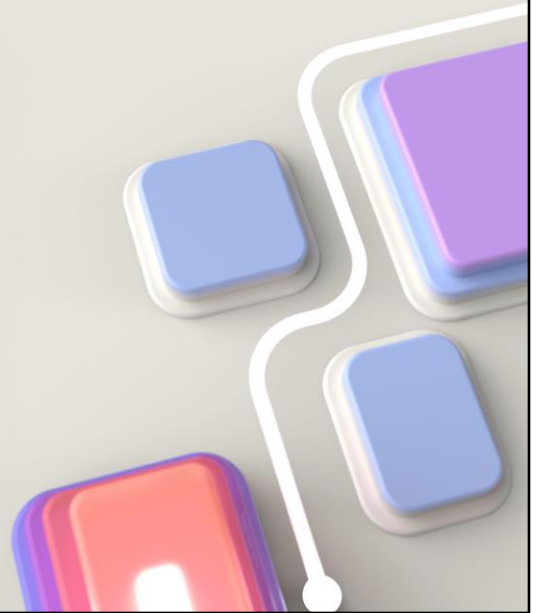
## Demo: Create a custom table

DEMO

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# Module 5: Create and manage columns in Microsoft Dataverse

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## Create columns in Dataverse

Data within Dataverse is stored within a set of records called tables.

In this module we will

- Define columns in Microsoft Dataverse
- Column types in Microsoft Dataverse
- Add a column to a table
- Create an auto numbering column
- Create an alternate key

Learning objectives

- Learn what a column is in Dataverse.
- Learn about the types of columns that are available in Dataverse.
- Add a column to a table.
- Create an auto-numbering column.
- Create an alternate key.

## Learning Objectives

After completing this module, you will be able to:

- 1 Learn what a column is in Dataverse.
- 2 Learn about the types of columns that are available in Dataverse.
- 3 Learn about choices.

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# Microsoft Learn module

## Create and manage columns within a table in Dataverse

<https://learn.microsoft.com/training/modules/create-manage-fields-within-entity>

## Working with choices in Dataverse

<https://learn.microsoft.com/training/modules/working-with-option-sets>



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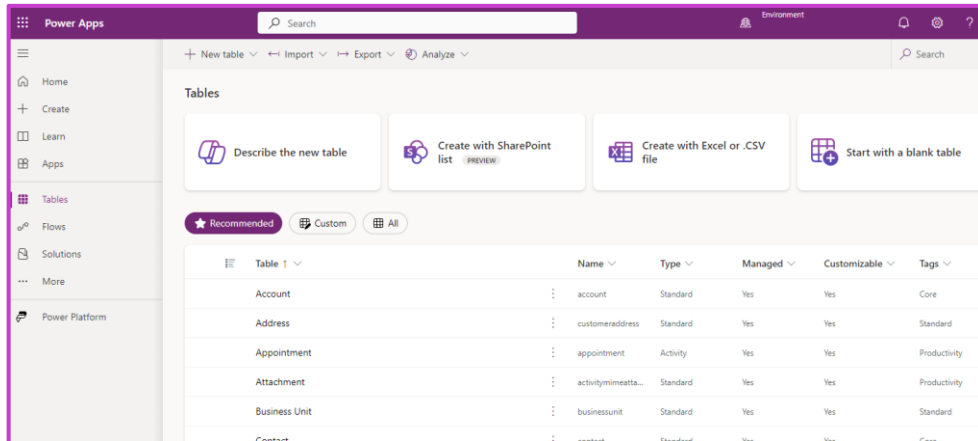
Link to student courseware on Learn

<https://learn.microsoft.com/training/modules/create-manage-fields-within-entity>

<https://learn.microsoft.com/training/modules/working-with-option-sets>

# Columns in Microsoft Dataverse

Columns are a way to store a discrete piece of information within a row in a table.



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Columns are a way to store a discrete piece of information within a record in a table. Columns have types, meaning that you can store data of a certain type in a field that matches that data type. For example, if you have a solution that requires dates, then you would store the date in a field with the type of Date. Similarly, if you want to store a number, then you store the number in a field with the type of Number.

The number of columns within a table varies from a few columns to a hundred or more. If you need more than a few hundred columns in a table, you might want to reconsider how you are structuring data storage for your solution because, likely, there is a better way.

Every database in Microsoft Dataverse starts with a standard set of tables and each standard table has a standard set of columns.

**Tip:** Always use standard tables and columns when possible. You can rename a table if that makes the table more understandable in the context of your solution. Always review the list of standard tables and make sure a standard table will not meet your needs before you create a new table.

# Add a column to a table

You can add columns when you create a new custom table, or you can add columns to an existing table at any time.

- Display name
- Descriptions
- Data type
- Format
- Behavior
- Required
- Searchable
- Schema name
- Advanced options

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## New column

Previously called fields. [Learn more](#)

Display name \*

Description ⓘ

Data type \* ⓘ

Format \*

Behavior ⓘ

Required ⓘ

☒ Searchable ⓘ

Advanced options ^

Schema name \* ⓘ

You can add columns when you create a new custom table, or you can add columns to an existing table at any time. Adding a new column is the same whether you are creating a new table or adding to an existing table.

**Tip:** Before you create a custom column, evaluate whether an existing column meets your requirement.

1. Sign in to Power Apps.
2. Select the **Dataverse** option on the left-hand menu.
3. Open an existing table or create a new table.
4. Select the dropdown by **+ New** and then **Column** on the menu at the top of the page.
5. Enter information in the following columns:
  - a. **Display name** - The name that is shown to users.
  - b. **Description** - This is a description of the column.
  - c. **Data type** - This is the type of data that you want to store in the column.
  - d. **Format**- The display format of the data to be stored in the column.
  - e. **Behavior**- Specifies whether the field will be a simple, calculated, or rollup field
  - f. **Required** - Select this check box if you want to ensure that this column always has a value when a user tries to add a record to this table.
  - g. **Searchable** - Clear this check box for columns for the table that you do not use. When a column is searchable, it appears in **Advanced Find** and is available when you are customizing views. Clearing this



check box reduces the number of options that are shown to people who are using **Advanced Find**, and helps make it easier for users to create custom views without seeing unused columns. Clear the **Searchable** check box when you are working with columns in a standard or complex table that you do not use.

- h. **Schema name**- This is the internal name that is used by your application.
- i. **Maximum character count** - Use to define the maximum length of the data that a user can enter in this column. Note that this is used with text columns.
- j. **Minimum and Maximum Values** - This column is available and used with number columns.
- k. **Size of number**- This column is available and used with number columns. It specifies whether the format will be int or bigint, which has implications on rollup

## Column data types

1 Single line of text

6 Currency

11 Choices

2 Multiple lines of text

7 Date and time

12 Yes/No

3 Whole number

8 Lookup

13 File

4 Floating point number

9 Customer

14 Image

5 Decimal number

10 Choice

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**Single Line of Text** - A text value intended to be displayed in a single-line text box.

- **Text**
- **Text Area** - A text value intended to be displayed in a multi-line text box. If you require more than 4,000 characters, use a Multiline Text data type.
- **Email** - A text value that is validated as an email address and rendered as a mailto link in the column.
- **URL** - A text value that is validated as a URL and rendered as a link to open the URL.
- **Ticker Symbol** - A text value for a ticker symbol that will display a link that will open to show a quote for the stock ticker symbol.
- **Phone** - A text value that is validated as a phone number rendered as a link to initiate a phone call by using Skype.
- **Whole Number** - A number value presented in a text box.
- **None**
- **Duration** - A number value presented as a drop-down list that contains time intervals. A user can select a value from the list or type an integer value that represents the number of minutes.
- **Language code** - A number value presented as a drop-down list that contains a list of languages that have been enabled for the environment. If no other languages have been enabled, the base language will be the only option. The saved value is the Locale Identifier (LCID) value for the language.
- **Timezone** - A number value presented as a drop-down list that contains a list of time zones.
- **Date and Time** - A date and time value.
- **Date Only** - A date and time value that only displays a date. The time value is stored as 12:00 AM (00:00:00) in

the system.

- **Currency** - A money value for any currencies that are configured for the environment. You can set a level of precision, or you can choose to base the precision on a specific currency or a single standard precision that is used by the organization.
- **Decimal Number** - A decimal value with up to 10 points of precision. See [Using the right type of number](#) for more information.
- **Floating Point Number** - A floating point number with up to 5 points of precision. See [Using the right type of number](#) for more information.
- **Image** - Displays a single image for each record in the application. Each table can have one image column. The name that you enter when creating an image column will be ignored. Image columns are always named **tableImage**. You can only have one image column for each table.
- **Lookup** - Creates a reference to a single row for a single target row type.
- **Choices** - Displays a list of options where more than one option can be selected.
- **Multiline Text** - A text value intended to be displayed in a multi-line text box. This value is limited to a maximum of 1,048,576 characters. You can also set a lower Max Length.
- **Choice** - Displays a list of options where only one can be selected.
- **Yes/No** - Displays two options where only one can be selected. You can choose which labels are displayed for each option. The default values are Yes and No.

**Tip:** You can add any combination of columns to a custom or standard table to meet your needs, but you can't delete a standard column from a standard table.

## Other column details

- |   |                        |   |             |
|---|------------------------|---|-------------|
| 1 | Column display formats | 6 | Auditing    |
| 2 | Column properties      | 7 | Description |
| 3 | Column requirement     |   |             |
| 4 | Searchable             |   |             |
| 5 | Column security        |   |             |

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Note: Audit is enabled by default

- Display name** - The name that is shown to users.
- Data type** - This is the type of data that you want to store in the column.
- Required** - Select this check box if you want to ensure that this column always has a value when a user tries to add a record to this table.
- Searchable** - Clear this check box for columns for the table that you don't use. When a column is searchable, it appears in **Advanced Find** and is available when you're customizing views. Clearing this check box reduces the number of options that are shown to people who are using **Advanced Find**, and helps make it easier for users to create custom views without seeing unused columns. Clear the **Searchable** check box when you're working with columns in a standard or complex table that you don't use.
- Behavior** - Choose between Simple or Calculated, and use this to define a calculation or a rollup in this column.
- Description** - This is a description of the column.
- Maximum character count** - Use to define the maximum length of the data that a user can enter in this column. This is used with text columns.
- Minimum and Maximum Values** - This column is available and used with number columns.

# Date and Time data type

- Dates stored in UTC
- Date and Time or Date Only
- Time zone adjustment
  - User Local
  - Date only
  - Time zone independent

Data type \* ⓘ  
Date and time

Format \* ⓘ  
Date only

Behavior ⓘ  
Simple

Required ⓘ  
Optional

☒ Searchable ⓘ

Advanced options ^

Schema name \* ⓘ  
dem\_

Time zone adjustment \* ⓘ  
User local

Date only  
User local  
Time zone independent

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# Choice data type

Choices provide an exact list of values that people can choose.

- List of values
- Key/Value pairs
- Global vs Local

The screenshot shows a 'User Information' form with fields for Home Phone, Other Phone, Email 2, Preferred Phone, Mobile Alert Email, and Pager. The 'Preferred Phone' field is a dropdown menu with a list of options: --Select--, --Select--, Main Phone, Other Phone, Home Phone, and Mobile Phone. The 'Mobile Alert Email' field is also a dropdown menu with a list of options: --Select--, Main Phone, Other Phone, Home Phone, and Mobile Phone.

The 'New choice' dialog box shows a 'Display name' field with the value 'PC Type'. Below it is a table of choices with columns for 'Label' and 'Value'. The table contains four rows: Laptop (930.620.000), Desktop (930.620.001), Tablet (930.620.002), and Server (930.620.003). There is a '+ New choice' button at the bottom.

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The 'Choice' configuration panel includes the following settings:

- Data type \***: Choice
- Behavior**: Simple
- Required**: Optional
- Searchable**: ☒ (with a sub-option 'Selecting multiple choices is allowed' which is unchecked)
- Sync with global choice? \***: ☒ Yes (recommended) (with a note: 'Can be used in multiple tables, and will stay updated everywhere.') and ☐ No (with a note: 'Creates a local choice that can only be used in one table. People using it can add new choices.')
- Sync this choice with \***: (empty dropdown)
- Default choice \***: None

Choices provide an exact list of values that people can choose. Choices help streamline new record creation and standardize data entry, and they are best suited for a small set of standard short terms rather than a long list of complex entries. You should not use choices when the field usually contains unique entries that are not easily standardized.

For example, consider the use of choices on a sales tracking page. A choice is well-suited to capture the stage of a sales lead if you have a short list of possible sales stages. On the other hand, choices are not a good option when you are entering a street name because, likely, there could be hundreds or thousands of street name choices in nearly any city.

**Tip:** Make sure that you monitor the quality of data when using a choice. As a best practice, check the data that you collect with a choice to ensure that users are not disproportionately using the default value rather than selecting the proper option within the choice. You might find that users accept the default so that they can enter data in the page without much thought. Determine what incentive users have to enter high quality data rather than the first entry or default in a choice, and avoid terms like "other" because options like these do not provide meaningful data.

Choices are managed as a separate list and then associated with a drop-down field. When used properly and closely managed, choices help ensure consistent data entry and improves the quality and usefulness of the data that you collect. choices can help create meaningful reports and identify trends and clusters of data. Finally, you can make fields of type choices required when they are created or edited.

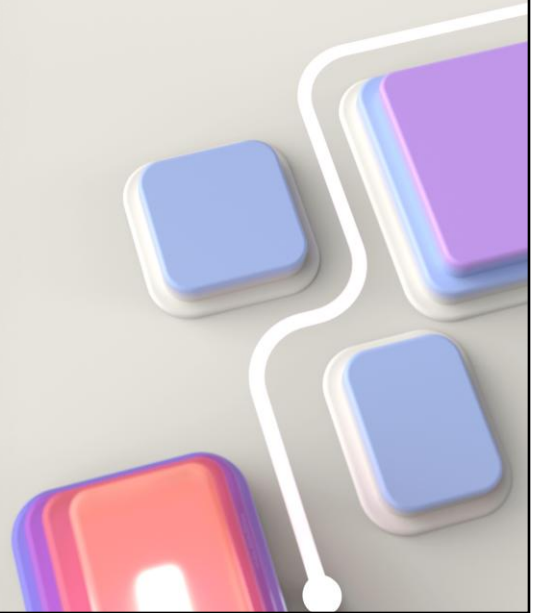
## Demo: Create a custom column

DEMO

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# Module 6: Create relationships between tables in Microsoft Dataverse

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## Create a relationship between tables in Dataverse

This module will show how and why you can separate data into tables and how to relate between tables to build complex and robust business solutions. It will also explain the different kinds of relationships that you can define between tables in Dataverse.

In this module we will:

- Relate one or more tables - Introduction
- Relationship types that are available in Microsoft Dataverse
- Create a one-to-many relationship between tables
- Create a many-to-many relationship between tables
- Edit or delete relationships

### Learning objectives

- Why you should segment data that is used by your solutions into many tables.
- Why you need to relate one table to another.
- How to build relationships between tables.
- How to select the proper relationship type when you are building solutions with Dataverse.



## Learning Objectives

After completing this module, you will be able to:

- 1 Why you should segment data that is used by your solutions into many tables.
- 2 Why you need to relate one table to another.
- 3 How to build relationships between tables.

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# Microsoft Learn module

## Create a relationship between tables in Dataverse

<https://learn.microsoft.com/training/modules/create-relationship-between-cds-entities>



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Link to student courseware on Learn

<https://learn.microsoft.com/training/modules/create-relationship-between-cds-entities>

# Table relationships

Table relationships define the different ways table records can be associated with records from other tables or the same table.

- **One-to-many relationships:** In a one-to-many (1:N) table relationship, many related table records are associated with a single primary table record in a parent/child relationship.
- **Many-to-many relationships:** In a many-to-many (N:N) table relationship, many table records are associated with many other table records. Records that are related through N:N table relationships are considered peers.
- When I delete a record, should any records that are related to that record also be deleted?
- When I assign a record to a new owner, do I also have to assign all related records to the new owner?
- How can I streamline the data entry process when I create a new related record in the context of an existing record?
- How should people who view a record be able to view the related records?

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Table relationships define the different ways table records can be associated with records from other tables or the same table. Table relationships are metadata. They let queries retrieve related data very efficiently. Use table relationships to define the formal relationships that define the table or that most records can use.

When you look at the solution explorer you might think that there are three types of table relationships but actually there are only two, see below:

- **One-to-many relationships:** In a one-to-many (1:N) table relationship, many related table records are associated with a single primary table record in a parent/child relationship.
- **Many-to-many relationships:** In a many-to-many (N:N) table relationship, many table records are associated with many other table records. Records that are related through N:N table relationships are considered peers.

The N:1 (many-to-one) relationship type exists in the user interface because the designer shows you a view grouped by tables. 1:N relationships actually exist between tables and refer to each table as either a Primary/Current table or Related table. The related table, sometimes called the child table, has a lookup column that allows storing a reference to a record from the primary table, sometimes called the parent table. A N:1 relationship is just a 1:N relationship viewed from the related table.

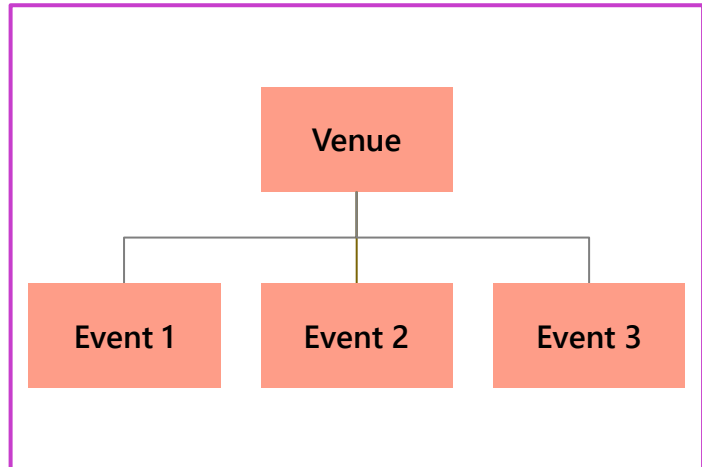
Besides defining how records can be related to other records, 1:N table relationships also provide data to address the following questions:

- When I delete a record, should any records that are related to that record also be deleted?
- When I assign a record to a new owner, do I also have to assign all related records to the new owner?

- How can I streamline the data entry process when I create a new related record in the context of an existing record?
- How should people who view a record be able to view the related records?

## Relationship types – One-to-many (1:N)

- One record of a primary table
- Zero, one or many records of a related table
- Related table has a lookup column
- N:1 is a reverse 1:N relationships



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One-to-many (1:N) relationships are the most common type of the relationships found in Dataverse for Apps.

For example, *one* venue can hold multiple events, or one company can have multiple employees.

Each one-to-many relationship has the following characteristics:

- One record of a *primary* table
- Zero, one, or many records of a *related* table
- Related table has a *lookup* column to store unique identifier of a primary table.
- N:1 is not a separate type, it is a 1:N relationship viewed from the other side

## Relationship behaviors



**Preserve data integrity**



**Rollup activity view**



**Parental relationships**



**Automate business processes**

Advanced options

Type of behavior \* ⓘ

Custom

Delete \*

Cascade All

Cascade All

Remove Link

Restrict

Assign \*

Cascade All

Unshare \*

Cascade All

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Some tables do not make sense on their own and they are typically created to support other tables and processes. They will typically have a required lookup column to link to the primary table they support. For example, business requirement could be that Project record should not exist without a related account. In this case Project will have an N:1 relationship with the Account and the account lookup column will be set to required.

What should happen when the primary record is deleted? You can use the relationship behavior to define this according to the rules for your business. Two options are:

- Prevent deleting the primary table so that the related table records can be reconciled, perhaps by associating them with a different primary table.
- Allow the related tables to be deleted automatically with the deletion of the primary table record.

If the related table doesn't support a primary table, i.e. lookup column is not set to required, you can allow the primary table to be deleted and the value of the lookup will be cleared.

# Mappings

- Operate on one-to-many relationships
- One-time only copy
- Used in model-driven app forms when creating a row from a sub-grid in the form
- Can use the `InitializeFromRequest` message in the API

**Create Field Mapping From Account to Contact**  
Create field mapping.

Source Entity Fields

| Name ↑                   | Display Name   | Ty   |
|--------------------------|----------------|------|
| accountcategorycode      | Category       | Opti |
| accountclassification... | Classification | Opti |
| accountid                | Account        | Prim |
| accountnumber            | Account Num... | Sing |
| accountratingcode        | Account Rating | Opti |

1 - 142 of 142 (0 selected)

Target Entity Fields

| Name ↑             | Display Name     | Ty    |
|--------------------|------------------|-------|
| accountrolecode    | Role             | Opti  |
| address1_addressid | Address 1: ID    | Prim  |
| address1_fax       | Address 1: Fax   | Sing  |
| address1_latitude  | Address 1: La... | Float |
| address1_longitude | Address 1: Lo... | Float |

1 - 134 of 134 (0 selected)

Help OK Cancel

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You can map columns between tables that have a relationship. This lets you set default values for a record that is created in the context of another record.

- In model-driven apps mappings are used when creating a related record from within the form or another record which has a 1-to-many relationship
- Values in columns on the parent record are copied to the child record.
- Only columns of the same data type can be mapped.
- Users can overwrite the values before the record is created
- The copy of values in once only and no future synchronization will take place.

Note: Use the Classic solution explorer to add mappings to the relationship.

The `InitializeFromRequest` request can be used in code to create one record from another and to prepopulate the record

<https://learn.microsoft.com/power-apps/maker/data-platform/map-entity-fields>

<https://learn.microsoft.com/power-apps/developer/data-platform/customize-entity-attribute-mappings>

<https://learn.microsoft.com/power-apps/developer/data-platform/webapi/reference/initializefrom?view=dataverse-latest>

## Demo: Create One-to-many relationship

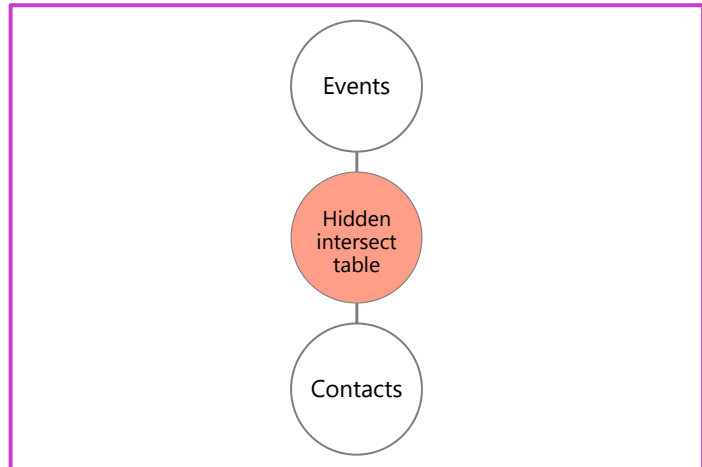
DEMO

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## Relationship types – Many-to-many (N:N)

- **Many-to-many (N:N) is symmetrical**
- **Hidden intersect table created**
- **Two One-to-many relationships**
- **Intersect table cannot be customized**
  - Can be used by report designers and developers



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1:N (one-to-many) table relationships establish a hierarchy between records. With N:N (many-to-many) relationships there is no explicit hierarchy. There are no lookup columns or behaviors to configure. Records created using N:N relationships can be considered peers and the relationship is reciprocal.

For example, for the companies (Accounts) you may want to record information about the board members (Contacts). One company may have many board members while each contact can be on the boards of multiple companies.

With N:N relationships a special hidden table is created called a Relationship (or intersect) table. This table has a N:1 relationship with each of the related tables and only stores the necessary values to define the relationship. You can't add custom columns to a relationship table, and the primary use of the intersect table is to link records for the reporting purposes.

In the app, the operations of adding or removing records from the relationship are called *associate* and *disassociate*, respectively.

## Relationship types – N:N (connections)

Defines a connection between table rows without creating a formal relationship

### Connection roles:

- Name
- Connection role category
- Description

The screenshot shows the 'New Connection' form in Dataverse for Apps. The form is titled 'CONNECTION New Connection' and has a 'Connected From' field set to 'Maria Campbell (sample)'. Below this, there is a 'Connect To' section with a 'Details' tab. The 'Details' tab shows a table with two rows: 'Start Date' (1/01/2015) and 'End Date' (1/01/2018). The 'As this role' field is set to 'Former Employee'.

| Connected From          | As this role    |
|-------------------------|-----------------|
| Maria Campbell (sample) | Former Employee |

| Start Date | End Date  |
|------------|-----------|
| 1/01/2015  | 1/01/2018 |

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Creating relationships requires planning and administrative effort. With Dataverse for Apps you can define connections between table records without creating a table relationship. By using connections users can establish a named link between records to establish a less formal relationship which doesn't justify creating an actual table relationship. For example, connections between accounts and contacts may include employees, board members, shareholders, customers, suppliers, etc. Some connections can also be reciprocal, such as child and parent, husband and wife, or doctor and patient.

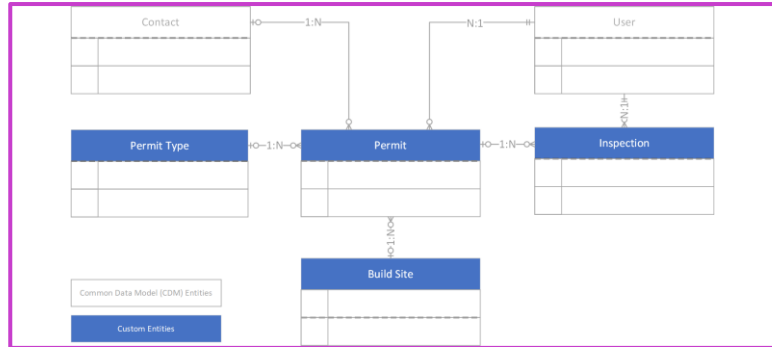
Connections are a way for users to record associations between almost any two records and provide additional information to describe how the records are linked to one another.

# Learning Path 1 practice labs



## Data Modeling

### • Lab 1. Data modelling

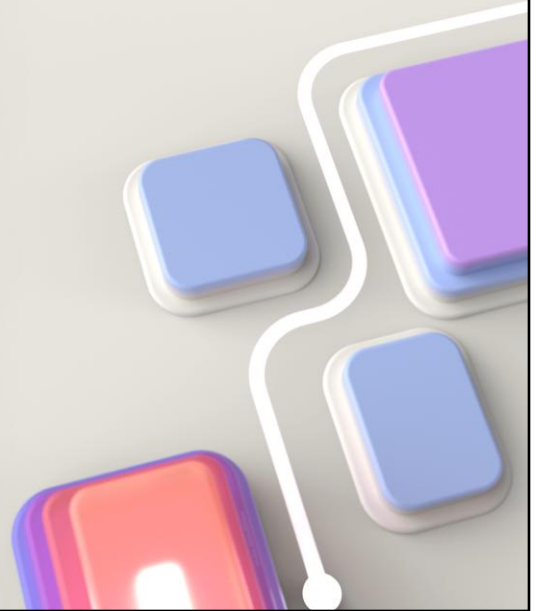


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A regional building department issues and tracks permits for new buildings and updates for remodeling of existing buildings. Throughout this course you will build applications and perform automation to enable the regional building department to manage the permitting process. This will be an end-to-end solution which will help you understand the overall process flow.

# Module 7: Create and define calculated and rollup columns in Microsoft Dataverse

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## Create and define calculation or rollup columns in Dataverse

This module shows you how to build calculations or rollups that are defined and triggered within the data in Dataverse, regardless of the app that accesses that data set.

In this module we will

- Introduction to rollup columns
- Create a rollup column
- Introduction to calculation columns
- Create a calculation column

### Learning objectives

- Define a rollup column.
- Create a rollup column.
- Identify a calculation column.
- Create a calculation column.

## Learning Objectives

After completing this module, you will be able to:

- 1 Define a rollup column.
- 2 Create a rollup column.
- 3 Identify and create a calculation column.

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# Microsoft Learn module

## Create and define calculation or rollup columns in Dataverse

<https://learn.microsoft.com/training/modules/create-define-calculation-rollup-fields>



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Link to student courseware on Learn

<https://learn.microsoft.com/training/modules/create-define-calculation-rollup-fields>

# Calculated and rollup columns

## Simple column:

Contains data not based on formula

Example: First name, birthday, or email address

## Calculated column:

Contains calculations that use columns from the current table or related parent table

Example: Full name (= First name + last name)

## Rollup column:

Contains an aggregate value computed from the rows related to a row in a one-to-many relationship

Example: Potential revenue (= Sum of revenues of all open opportunities for a given account)

Behavior ⓘ

Simple

Simple

Calculated

Rollup

Searchable ⓘ

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Behavior option when creating column.

Cannot change after save

## Calculated columns

String, Numeric, and DateTime functions available using Power Fx formula columns

## Rollup columns

Rollup columns perform calculations on values that are stored in a column in a related table across a one-to-many relationship. Rollup columns can calculate sum, min, max, and count values in a fashion that is similar to how these same functions are used within Microsoft Excel. Rollup columns are read-only and can calculate values based on a system account with full access to all rows in a related table, so they might calculate values based on rows that a current user cannot access due to security settings that prevent the user from access.

Rollup columns are calculated by Microsoft Dataverse and the value of the rollup calculation is stored within the column in the table, meaning that rollup columns can be used for sorting, searching, and filtering like any other column.

Rollup calculations are calculated asynchronously (meaning not in real time) by using a scheduled system job in Dataverse. Rollup calculations run once an hour by default,

When you configure a rollup column, two recurring jobs will be created. These two jobs are a **Mass Calculate Rollup Column** and a **Calculate Rollup Column**. The **Mass Calculate Rollup Column** will happen, by default, 12

hours after you configure the rollup column. This will recalculate the rollup column for all existing rows that contain the column. The **Calculate Rollup Column** job will occur every hour, and is an incremental calculation of the rollup column. This means that after the **Mass Calculate Rollup Column** executes, the **Calculate Rollup Column** job will only have to update the column to include rows that have since been created, updated or deleted.



# Calculated and rollup column data types

## Calculated column

- Single line of text
- Choice
- Yes/No
- Whole number
- Decimal number
- Currency
- Date Time

## Rollup column

- Whole number
- Decimal number
- Currency
- Date Time

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Creating a calculated column is a simple process; however, you should consider the following points before you begin:

You must create a supported column type and then define the calculation. The following columns support calculations:

- Single line of text
- Choice
- Yes/No
- Whole number
- Decimal number
- Currency
- Date Time

Creating a rollup column is a simple process, but you should consider the following points before you begin.

You must create a supported column type in a table and then define the rollup that you want to calculate. The following column types support rollups:

- Whole Number
- Decimal Number

- Currency
- Date Time

Rollups can be filtered and can rollup through hierarchical relationship

# Calculated and rollup column limitations

## Calculated columns

- Calculated on retrieve
- Can use columns on table and columns in many-to-one relationships

## Rollup columns

- Max 100 per environment
- Max 10 per table
- Recalculated every hour
- Requires a one-to-many relationship

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### Calculated columns

- Calculated columns can span two tables only.
- A calculated column can contain a column from another table (spanning two entities: current table and parent row).
- A calculated column can't contain a calculated column from another table that also contains another column from a different table (spanning three entities).

Sorting is disabled on:

- A calculated column that contains a column of a parent row.
- A calculated column that contains a logical column (for example, address column).
- A calculated column that contains another calculated column.

### Rollup columns

- Rollups calculate a value from a column in one or many child rows in a related table.
- You are limited to a maximum of ten (10) rollups for each parent table.
- Because rollup columns persist in the database, they can be used for filtering or sorting just like regular columns.
- Remember that rollups are asynchronous and are calculated by Dataverse on a set schedule, so you might not see the expected value in a rollup until that job runs.

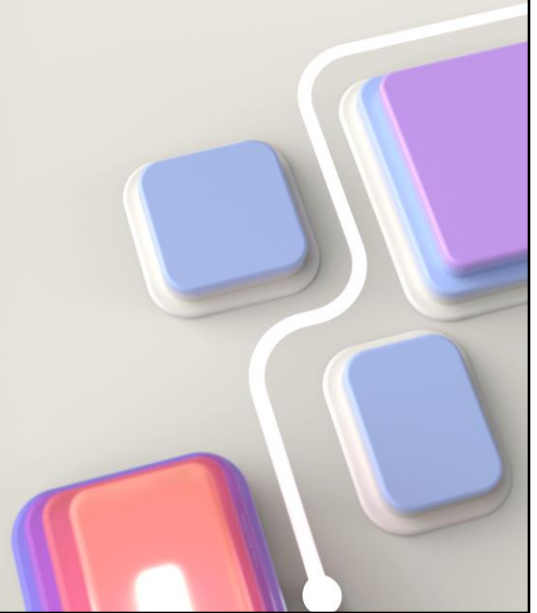
## Calculated and rollup column limitations

- 1 Values are read-only
- 2 Values not audited
- 3 Do not trigger workflow, code, or Power Automate cloud flows
- 4 Calculated columns cannot contain a calculated column from another table that also contains another column from a different table (spanning three tables)

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# Module 8: Define and create business rules in Microsoft Dataverse

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## Define and create business rules in Dataverse

The ability to build business rules that ensure consistent business logic whatever app is accessing that data set is imperative to a successful business operation.

This module will show you how you can build business rules that are triggered anytime they're used within Dataverse.

In this module we will

- Define business rules - Introduction
- Define the components of a business rule
- Create a business rule

Learning objectives

- Define business rules in Dataverse.
- Create and manage business rules in Dataverse.

## Learning Objectives

After completing this module, you will be able to:

- 1 Define business rules in Dataverse.
- 2 Create and manage business rules in Dataverse.

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# Microsoft Learn module

## Define and create business rules in Dataverse

<https://learn.microsoft.com/training/modules/define-create-business-rules>



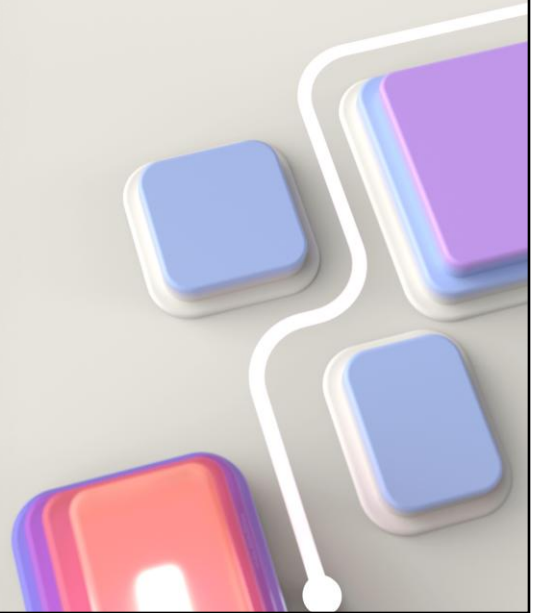
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Link to student courseware on Learn

<https://learn.microsoft.com/training/modules/define-create-business-rules>

# What behavior is expected when the logic happens on the server?

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## Discussion

Engage students in a discussion about logical expectations of changes that happen locally vs at the server. This will help them understand more about the expectations of the logic they will be building.

Talk about things like scope of business rules, that if scope is set to a specific form only in a model driven app, that if that same table is used in a canvas app, the logic won't be applied to the user's actions in the canvas app.



# Business rules

**You can create business rules and recommendations to apply logic and validations without writing code**

- 1** Automate and enforce common requirements
- 2** Rules are created in the context of a table
- 3** Rules can execute on both client and server
- 4** Provide an alternative to developer code-based rule implementation

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You can create business rules and recommendations to apply logic and validations without writing code.

They can also be used to validate stored data or show error messages. Model-driven apps can use business rules to show or hide columns, enable or disable columns, and create recommendations based on business intelligence.

[!TIP] Business rules are usually defined for a table and apply to all forms, but you can define a business rule for a specific model-driven form. Canvas apps cannot have a business rule applied to a specific form.

Business rules give you a powerful way to enforce rules, set values, or validate data regardless of the form that is used to input data. Additionally, business rules are effective in helping to increase the accuracy of data, simplify application development, and streamline the forms presented to end users.

Business rules provide a simple interface to implement and maintain fast-changing and commonly used rules.

The business rules defined for a table apply to both canvas apps and model-driven apps if the table is used in the app.

By combining conditions and actions, you can do any of the following with business rules:

- Set column values.
- Clear column values.
- Set column requirement levels
- Validate data and show error messages

# Define the components of a business rule

Business rules encapsulate logic in a predefined set of steps that will run each time that data is entered or modified, and the data meets the criteria to trigger the business rule

- 1 Condition – Business rules start with a condition
- 2 Action – Logic
- 3 Scope – Defines where a business rule is applied

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Business rules encapsulate logic in a predefined set of steps. The steps run each time data is entered or modified and the data meets certain criteria to trigger the business rule. The business rule will run whatever way that data was added or edited in the table. Additionally, business rules are server-side, meaning that the logic runs on the servers that manage Microsoft Dataverse.

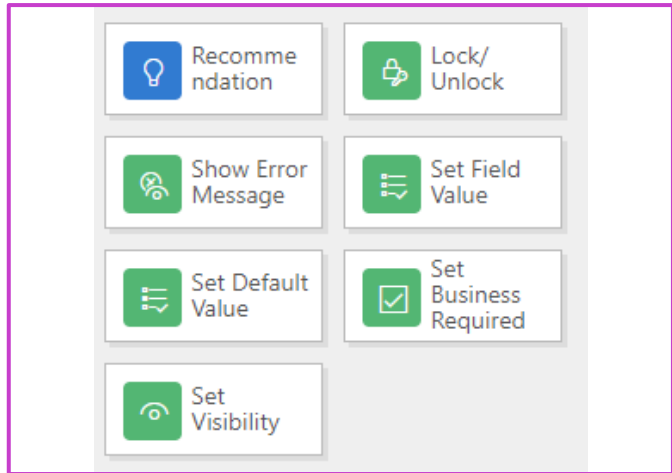
Business rules have the following components:

- **Condition** - All business rules have a condition. It's a trigger and is used to determine if the business rule is run based on the values that are added or edited in a table. A condition always runs as true or false, and you can have more than one condition in a business rule.
- **Action** - An action is some logic, like setting a column to a certain value, that runs on either the true or false branch of a condition.
- **Scope** - Where the rule is applied
  - Entity
  - All forms
  - Specific form

# Define the components of a business rule

## Actions

Set column values  
Clear column values  
Set column requirement levels  
Show or hide columns  
Enable or disable columns  
Validate data and show error messages  
Create business recommendations



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By combining conditions and actions, you can do any of the following with business rules:

- Set column values
- Clear column values
- Set column requirement levels
- Show or hide columns
- Enable or disable columns
- Validate data and show error messages
- Create business recommendations based on business intelligence

# Define the components of a business rule

## Scope

A business rule only applies in a canvas app when the scope is set to **Entity**.

| Scope of a business rule                  | Applies to                        |
|-------------------------------------------|-----------------------------------|
| Entity (Table)                            | Model-driven app forms and server |
| All forms                                 | All model-driven app forms        |
| Specific form (Account form, for example) | Just that model-driven app form   |

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**Scope** - Business rules can have a scope

- Entity
- All forms
- Specific form

[!TIP] A business rule that is used by a canvas app always has the scope set to **Entity/Table**.

## Differences between canvas and model-driven apps for business rules

**Model-driven apps can use all actions available on business rules, however not all business rule actions are available for canvas apps**

- 1** Show or hide columns
- 2** Enable or disable columns
- 3** Create business recommendations based on business intelligence

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Model-driven apps can use all actions available on business rules, however not all actions are available for canvas apps.

Model-driven apps can use business rules to:

- Show or hide columns (model-driven apps only).
- Enable or disable columns (model-driven apps only).
- Create business recommendations based on business intelligence (model-driven apps only).

These actions are not available on Canvas apps

## Demo: Business rules

DEMO

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Show how to use the business rule builder.

*Alternatively show video <https://youtu.be/lqEIX5MeLak>*

# Module 9: Manage security in Microsoft Dataverse

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## Dataverse security

**Note:** the JTA/OD restricts PL-400 to Configuring Microsoft Power Platform to support development. So covers the basics of the Dataverse security features.

This module will show you how you can set permissions to limit access to an environment or limit which users can view, edit, or delete data in an environment within Dataverse.

In this module we will

- Introduction to environment roles
- Understand environment roles
- Adding or disabling an environment user
- Understand security concepts in Dataverse
- Understand user security roles and security role defaults
- Check the roles that a user belongs to
- Configure Dataverse teams for security
- Configure Dataverse group teams for security

Learning objectives

- Learn about security roles and apply them to users in an environment.
- Learn how to add users to an environment.
- Understand security concepts in Dataverse
- Identify default security roles.
- Create a custom role.
- Create a custom security role and assign it to entities and users.

- Learn how to configure Dataverse teams for security
- Learn how to configure Dataverse group teams for security



## Learning Objectives

After completing this module, you will be able to:

- 1** Learn about security roles and apply them to users in an environment.
- 2** Understand security concepts in Dataverse
- 3** Create a custom security role and assign it to entities and users.

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# Microsoft Learn module

## Get started with security roles in Dataverse

<https://learn.microsoft.com/training/modules/get-started-security-roles>



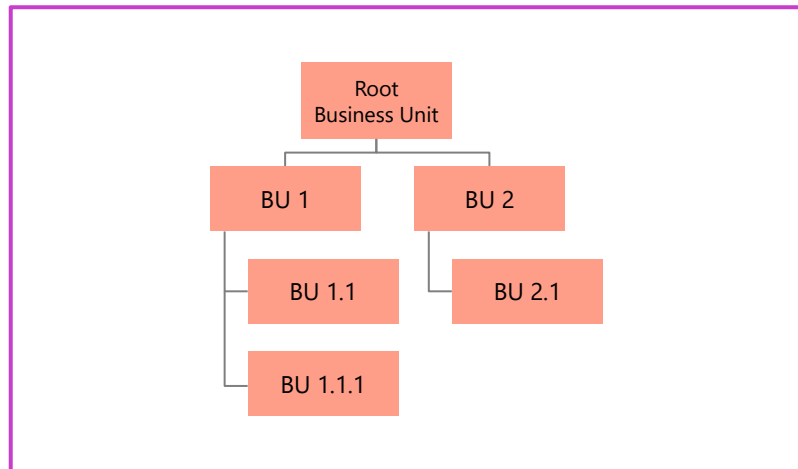
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Link to student courseware on Learn

<https://learn.microsoft.com/training/modules/get-started-security-roles>

## Security model building blocks

- Teams
- Business units
- Security roles
- Column-level security
- Access teams
- Hierarchy security



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One of the key features of Dataverse is its rich security model that can adapt to many business usage scenarios. This security model is only in play when there is a Dataverse data store in the environment. As an administrator, you likely won't be building the entire security model yourself but will often be involved in the process of managing users and making sure they have the proper configuration and troubleshooting security access related issues.

Data in Dataverse is securely stored so that users can see it only if you grant them access. Role-based security, based on the Dynamics 365 system allows you to control access to entities for different users within your organization.

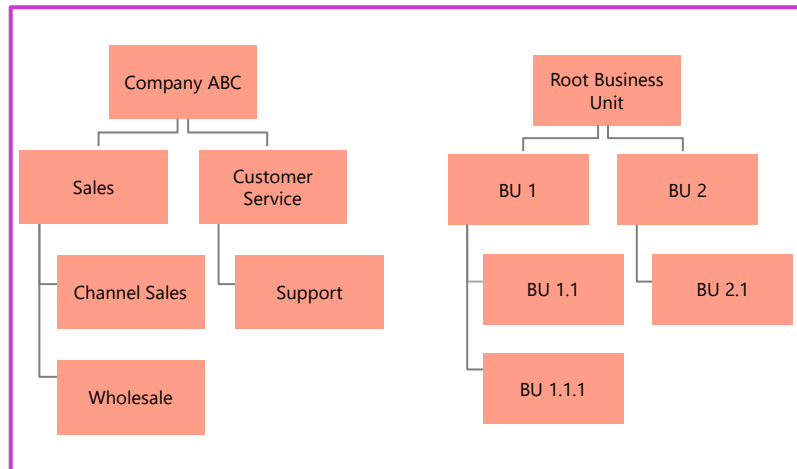
To configure an installation, you start by creating the Business Units and Security Roles that define the organizational structure. When Dataverse is deployed, the setup creates a root Business Unit that represents the primary organizational unit for the implementation that includes a set of default Security Roles.

Additional Security Roles can be configured to specify the privileges that will be granted to users who do different jobs, in different areas of the organization.

After the organization structure and security model is configured, users are created, and Security Roles are assigned to provide the privileges they require. Users can also be grouped into Teams that can be used to share records or personal views, charts and dashboards, and are also used to refine the security model.

## Business units

Security boundaries for users (and teams) for access and privileges to organizational data



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Business Units form a hierarchy that contains groups of Users and Teams, and the records the Users and Teams own. The hierarchy provides security boundaries to control the scope of the User permissions. This means that permissions can be granted to records so that users can perform different actions on records that are owned by other Users or Teams in their Business Unit from the actions that they can perform on records that are located in another Business Unit. Business Units are also helpful when you consider your reporting requirements. Depending on the access levels that are granted to Users to read records, sometimes a single report can be used by several users in different Business Units to receive the results that they must have. This occurs because the records that are reflected in the report are filtered to only include the records that the user has access to. If users have access to records for the whole Organization you can use Business Units in the queries of your report to filter or categorize results. When you plan a Dataverse deployment, it might be helpful to refer to an organization chart of the company structure. However, you do not have to replicate this in your Business Units. We recommend that you create only the Business Units that you must have to meet the security or reporting requirements of the business.

## Understand user security roles and security role defaults

- 1 **Security roles** are groups of permissions that you can assign to a user to grant them access and various capabilities and functionality like read, delete, or edit of records in a table within an environment
- 2 **Security roles are either:**
  - Standard security roles created with every instance of Microsoft Dataverse
  - Custom security roles created by an administrator

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Roles are groups of permissions that you can assign to a user to grant them access and various capabilities and functionality like read, delete, or edit of records in an entity within an environment. Roles are granular and can be assigned to one or many entities in an environment. Roles can also control certain actions like the ability to create a custom entity or option sets. Additionally, users are associated with one or many roles, and associating a user with a role gives them access to data and functionality that is specified within that role.

User security roles are either:

Standard and created with every instance of Microsoft Dataverse.

Custom and created by an administrator.

This unit examines each type of security role.

## Standard security roles

- 1 You can manage **Dataverse** security by using roles and then adding users to the environment and assigning roles to users
- 2 **The built-in roles available to assign to users:**
  - System Administrator
  - System Customizer
  - Environment Maker
  - Basic User
  - App Opener

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You can manage Dataverse security by using security roles and by adding users to the environment and assigning roles to users. A role has certain permissions that are associated with it, and you can associate a user with one or many roles.

The built-in roles are:

- System Administrator
- Environment Maker
- Basic User
- System Customizer
- App Opener

This unit examines each role to help you understand how it works within an environment.

The System Administrator role can perform all administrative actions on an environment, including the following tasks:

Add or remove a user or group from either the Environment Admin or Environment Maker role.

Provision a Microsoft Dataverse data store for the environment.

View and manage all resources that are created within an environment.

Set data loss prevention policies.

The System Customizer role has full permission to customize the environment. However, users with this role can only view records for environment entities that they create. More information: [Privileges required for customization](#)

The Environment Maker role can create resources within an environment such as apps, connections, custom connectors, gateways, apps that use Power Apps, and flows that use Power Automate. The following applies to members of the Environment Maker role:

Environment Makers can distribute the apps that they build in an environment to other users within an organization by sharing the app with individual users, security groups, or to all users in the organization.

The Basic User role Can run an app within the environment and perform common tasks for the records that they own. This only applies to non-custom entities.

Users or security groups can be assigned to either of these two roles by a System Administrator from the Power Platform Administration Center.

## Security roles overview

**Security Role: Account Manager (Preview)**

Details

Business unit: org0dc63f0b

When role is assigned to a Team  
Team member gets all team privileges by default.  
Team members can inherit team privileges directly based on access level. [Learn More](#)

Member's privilege inheritance  
Direct User (Basic) access level and Team privileges

Tables Miscellaneous privileges Privacy-related privileges

Show all tables

Compact Grid View ☒ On

| Table                           | Name                 | Record ownership | Permission Settin... | Create | Read         | Write | Delete |
|---------------------------------|----------------------|------------------|----------------------|--------|--------------|-------|--------|
| <b>Business Management (24)</b> |                      |                  |                      |        |              |       |        |
| Business Unit                   | businessunit         | Business Unit    | Reference            | None   | Organization | None  | None   |
| Channel Property Group          | channelpropertygroup | Organization     | No Access            | None   | None         | None  | None   |
| Currency                        | transactioncurrency  | Organization     | Reference            | None   | Organization | None  | None   |
| Document Template               | documenttemplate     | Organization     | No Access            | None   | None         | None  | None   |

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The security model in provides access to records and features of the application. Users can access records and perform actions on those records that are appropriate to their job function. A key objective of any security model is to prevent unauthorized access or modifications to data. In Dataverse, the security rights that you configure also have a direct effect on the way the system behaves and is displayed to the user. If a user does not have access to a specific table or application feature, references to the table or feature are not visible in the user's interface. By limiting access to only

the areas a user must have, you can help make the user experience easier. Dataverse uses a role-based security model to define the permissions that are suitable for a specific job role that is then assigned to Users or Teams. This model uses a combination of Business Units to define a hierarchy of Users and Teams, and Security Roles to define the privileges the Users and Teams should be granted.

The security permissions that are defined in each Security Role consist of *privileges* that describe the actions that are allowed at a selected *access level* that describe where the privilege applies. The permissions are displayed in the form by using a

series of icons, the key for which is included on the Security Role form as shown in the "Security Role Core Records Tab table Privileges"

Microsoft Dataverse has many standard default roles, but there might be times when you want to define a custom security role. Dataverse supports the following eight different record-level privileges that can be used to define how a user interacts with data for one or more tables for use in building a custom role. The available record-level privileges for custom roles include:

**Create** - Required to make a new record. The records that can be created depends on the access level of the permission that is defined in your security role.

**Read** - Required to open a record to view the contents. The records that can be read depends on the access level of the permission that is defined in your security role.

**Write** - Required to make changes to a record. The records that can be changed depends on the access level of



the permission that is defined in your security role.

**Delete** - Required to permanently remove a record. The records that can be deleted depends on the access level of the permission that is defined in your security role.

**Append** - Required to associate a record with the current record. For example, if a user has Append rights on an opportunity, the user can add a note to an opportunity. The records that can be appended depends on the access level of the permission that is defined in your security role.

**Append To** - Required to associate the current record with another record. For example, a note can be attached to an opportunity if the user has Append To rights on the note. The records that can be appended to depends on the access level of the permission that is defined in your security role.

**Assign** - Required to give ownership of a record to another user. The records that can be assigned depends on the access level of the permission that is defined in your security role.

**Share** - Required to give another user access to a record while keeping your own access. The records that can be shared depends on the access level of the permission that is defined in your security role.

These record-level privileges can be grouped as needed and associated with a custom role. That custom role can then be applied to one or many tables as needed.

[!TIP] Roles can be copied so you can quickly create similar roles that might be slightly different.

# Security roles in Power Platform admin center

Copy security role Rename security role Delete Save Save + close Configure column view

The Basic User role privileges cannot be adjusted due to its non-customizable nature. [Learn More](#)

Environments > PL400 > Settings > Security roles > Basic User

Search by table name or table privilege n...

Details

Business unit: org3c75f3e

When role is assigned to a Team  
Team member gets all team privileges by default. Team members can inherit team privileges directly based on access level. [Learn More](#)

Member's privilege inheritance  
Direct User (Basic) access level and Team priv...

Tables Miscellaneous privileges Privacy-related privileges

Show only assigned tables

Compact Grid View On

| Table                      | Name                   | Record owner... | Permission ... | Create | Read         | Write | Delete | Append | Append to    | Assign | Share        |
|----------------------------|------------------------|-----------------|----------------|--------|--------------|-------|--------|--------|--------------|--------|--------------|
| Business Management (17)   |                        |                 |                |        |              |       |        |        |              |        |              |
| Business Process Flows (3) |                        |                 |                |        |              |       |        |        |              |        |              |
| Core Records (44)          |                        |                 |                |        |              |       |        |        |              |        |              |
| ActivityMapper             | activitymapper         | Organization    | Reference      | None   | Organization | None  | None   |        |              |        |              |
| Account                    | account                | User or Team    | Custom         | User   | User         | User  | User   | User   | User         | User   | Organization |
| Action Card                | actioncard             | User or Team    | Custom         | User   | User         | User  | None   | User   | Organization | None   |              |
| Action Card User Settings  | actioncardusersettings | User or Team    | Private        | User   | User         | User  | User   |        |              |        | User         |
| Activity                   | activitypointer        | User or Team    | Custom         | User   | User         | User  | User   | User   | User         | User   | Organization |

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PPAC->Environment->Settings->Security Roles

# Privileges and access levels

## Privileges:

- Create
- Read
- Write
- Delete
- Append
- Append to
- Assign
- Share

## Access levels:

- User
- Business Unit
- Parent-Child Business Unit
- Organization

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**Create** - Required to make a new record. The records that can be created depends on the access level of the permission that is defined in your security role.

**Read** - Required to open a record to view the contents. The records that can be read depends on the access level of the permission that is defined in your security role.

**Write** - Required to make changes to a record. The records that can be changed depends on the access level of the permission that is defined in your security role.

**Delete** - Required to permanently remove a record. The records that can be deleted depends on the access level of the permission that is defined in your security role.

**Append** - Required to associate a record with the current record. For example, if a user has Append rights on an opportunity, the user can add a note to an opportunity. The records that can be appended depends on the access level of the permission that is defined in your security role.

**Append To** - Required to associate the current record with another record. For example, a note can be attached to an opportunity if the user has Append To rights on the note. The records that can be appended to depends on the access level of the permission that is defined in your security role.

**Assign** - Required to give ownership of a record to another user. The records that can be assigned depends on the access level of the permission that is defined in your security role.

**Share** - Required to give another user access to a record while keeping your own access. The records that can be shared depends on the access level of the permission that is defined in your security role.

## Types of teams

### Owner teams:

- Rows owned by groups of users instead of an individual user
- Number of teams is known at design time
- Reporting on progress by teams is required

### Access teams:

- Teams need to be dynamically formed and dissolved
- Number of teams is not known at design time
- Team members require different access rights to rows
- Some users require access to a single row without requiring ownership

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TEAM ▼

Managers ≡

General

|                 |                |
|-----------------|----------------|
| Team Name *     | HR Managers    |
| Business Unit * | HR             |
| Administrator * | Sven Mortensen |
| Team Type *     | Owner          |

Team members ▼

Search for records

Full Name ↑

Sven Mortensen

William Contoso

There are three types of Teams—owner Teams, access Teams and Azure AD Group team

Owner Teams can be assigned Security Roles and therefore own records,

Access Teams cannot have Security Roles and cannot own records, although records can be shared with the access Teams. Although an owner Team can be converted to an Access Team, you cannot do this the other way round. Access Teams are mainly intended for use with Access Team Templates.

## Entra ID group teams to manage user app and data access

- 1 A Entra ID group team, similar to an owner team, can own records and can have security roles assigned to the team.
- 2 The administration of app and data access for Microsoft Dataverse has been extended to allow administrators to use their organization's Entra ID groups to manage access rights for licensed Dataverse users.

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The administration of app and data access for Microsoft Dataverse has been extended to allow administrators to use their organization's Entra ID groups to manage access rights for licensed Dataverse users.

**Entra ID group team** is similar to owner teams, an Entra ID group team can own records and can have security roles assigned to the team. Security and Office are two group team types, and they correspond directly to Entra ID group types. Group security roles can be assigned only for a specific team or for a team member with user privileges that include members' privilege inheritance. Team members are dynamically derived (added and removed) when they access an environment based on their Entra ID group membership.

Using Entra ID groups lets administrators assign a security role with its respective privileges to all the members of the group, instead of having to provide the access rights to an individual team member.

Both types of Entra ID groups (Office and Security) can be used to secure user-access rights. Using groups lets administrators assign a security role with its respective privileges to all the members of the group, instead of having to provide the access rights to an individual team member.

Both types of Entra ID groups, Office and Security, with a Membership type Assigned can be used to secure user-access rights.

Membership type Dynamic User and Dynamic Device are not supported.

The administrator can create Entra ID group teams that are associated to the Entra ID groups in each of the Dataverse environments and assign a security role to these group teams. For each Entra ID group, the administrator can create group teams based on the Entra ID group membership types. The administrator can create separate group teams for owners, members, guests and members, and guests, and assign a respective security role to each of these teams.

When members of these group teams access these environments, their access rights are automatically granted based on the group team's security role.

Note: You can assign security roles directly to owner teams and Entra ID group teams and users. The environment picker only recognizes users who are members of Entra ID group teams and users who have security roles assigned to them directly.

## Create and manage column security profiles

- 1 Configure column-level security when a table contains certain columns that are more sensitive than the other data
- 2 Available for most out-of-the-box tables and custom columns
- 3 **To implement column-level security, a system administrator performs the following tasks:**
  - Enable column security on one or more columns for a table
  - Create and configure column security profiles
  - Associate one or more existing security profiles to grant access to specific users or teams

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Column-level permissions are granted at the table level, but you may have certain columns associated with a table that contain data that is more sensitive than the other columns. For these situations, you use column-level security to control access to specific columns.

The scope of column-level security is organization-wide and applies to all data access requests including the following:

Data access requests from within a client application, such as web browser, mobile client, or Microsoft Dynamics 365 for Outlook.

Web service calls using the Microsoft Dataverse web services (for use in plug-ins, custom workflow activities, and custom code)

Reporting (using Filtered Views)

Column-level security is available for the default columns on most out-of-the-box tables, custom columns, and custom columns on custom tables. Column-level security is managed by the security profiles.

To implement column-level security, a system administrator performs the following tasks.

Enable column security on one or more columns for a given table.

Associate one more existing security profile, or create one or more new security profiles to grant the appropriate access to specific users or teams.

A security profile determines the following:

Permissions to the secure columns

Users and Teams

A security profile can be configured to grant user or team members the following permissions at the column level:

Read. Read-only access to the column's data.

Create. Users or teams in this profile can add data to this column when creating a record.

Update. Users or teams in this profile can update the column's data after it has been created.

A combination of these three permissions can be configured to determine the user privileges for a specific data column.

[!IMPORTANT] Unless one or more security profiles are assigned to a security enabled field, only users with the system administrator security role will have access to the field.



# Summary



- Dataverse uses tables, columns, and relationships to help you build powerful, scalable data solutions
- Power Platform supports multiple environments each with a Dataverse data store
- Solutions are used to manage the customization of Dataverse components
- Dataverse support standard, activity, and virtual tables
- Dataverse tables have properties that affect the capabilities of the table in apps
- Dataverse has many column data types
- Dataverse supports one-to-many and many-to-many relationships
- Calculated and rollup columns remove the need for code to calculate values
- Business rules can perform business logic on model-driven app forms and at the platform level
- Access to table rows in Dataverse is managed using business units, teams, and security roles
- Access to table columns can be restricted by using column-level security

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## Modules

1. Introduction to Microsoft Dataverse
2. Manage environments
3. Manage customizations with solutions
4. Create and manage tables in Microsoft Dataverse
5. Create and manage columns in Microsoft Dataverse
6. Create relationships between tables in Dataverse
7. Create and define calculated and rollup columns in Dataverse
- 8. Define and create business rules in Microsoft Dataverse**
9. Manage security in Microsoft Dataverse



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